1)Address

Address.cpp

```
#include "Address.h"
#include <cstring>

Address::Address()
{
   houseNo = 0;
   pincode = 0;
   colony[0] = '\0';
   area[0] = '\0';
   city[0] = '\0';
}

Address::Address(int h, char c[], char a[], char ci[], int p)
{
   houseNo = h;
   pincode = p;

   strcpy(colony, c);
   strcpy(area, a);
   strcpy(city, ci);
}

void Address::accept()
{
   cout<<"Enter House No: ";
   cin>>houseNo;

   cout<<"Enter Colony: ";
   cin>>colony;
```

```
    cout<<"Enter Area: ";
    cin>>area;

    cout<<"Enter City: ";
    cin>>city;

    cout<<"Enter Pincode: ";
    cin>>pincode;
}

void Address::display()
{
    cout<<"\nHouse No: "<<houseNo;
    cout<<"\nColony: "<<colony;
    cout<<"\nArea: "<<area;
    cout<<"\nCity: "<<city;
    cout<<"\nPincode: "<<pincode<<endl;
}

bool Address::isSame(Address a)
{
    if(houseNo==a.houseNo &&
      strcmp(colony,a.colony)==0 &&
      strcmp(area,a.area)==0 &&
      strcmp(city,a.city)==0 &&
      pincode==a.pincode)
       return true;
    else
       return false;
}
```

```
bool Address::operator==(Address a)
{
   if(houseNo==a.houseNo &&
      strcmp(colony,a.colony)==0 &&
      strcmp(area,a.area)==0 &&
      strcmp(city,a.city)==0 &&
      pincode==a.pincode)
       return true;
   else
       return false;
}
```

Address.h

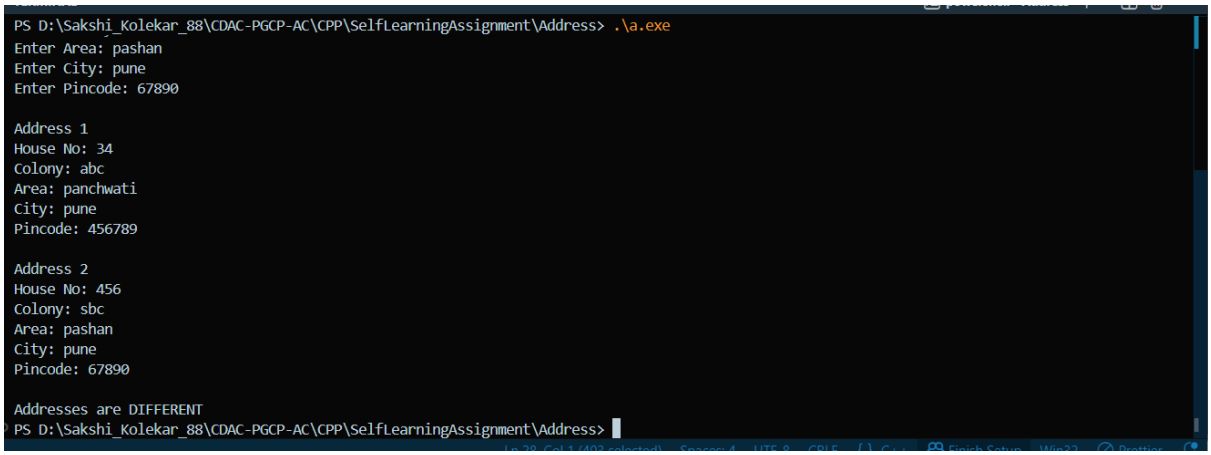
```
#ifndef ADDRESS_H
#define ADDRESS_H

#include <iostream>
using namespace std;

class Address
{
private:
   int houseNo;
   char colony[50];
   char area[50];
   char city[50];
   int pincode;

public:
   Address();
   Address(int, char[], char[], char[], int);
```

```
    void accept();
    void display();

    bool isSame(Address);
    bool operator==(Address);
};

#endif
```

Main.cpp

```
#include "Address.h"

int main()
{
    Address a1,a2;

    cout<<"Enter Address 1\n";
    a1.accept();

    cout<<"\nEnter Address 2\n";
    a2.accept();

    cout<<"\nAddress 1";
    a1.display();

    cout<<"\nAddress 2";
    a2.display();

    if(a1==a2)
        cout<<"\nAddresses are SAME";
```

```
    e
        cout<<"\nAddresses are DIFFERENT";

    return 0;
}
```



2)Main.cpp

```
#include "Mobile.h"
#include <cstring>

int main()
{
    MobilePhone m[50];
    int count=0;
    int choice;
    char search[30];

    do
    {
        cout<<"\n1 Add Mobile";
        cout<<"\n2 Display Mobiles";
        cout<<"\n3 Find Quantity of Model";
```

```
cout<<"\n4 Check Availability";
cout<<"\n5 Exit";

cout<<"\nEnter Choice: ";
cin>>choice;

switch(choice)
{
   case 1:
      m[count].accept();
      count++;
      break;

   case 2:
      for(int i=0;i<count;i++)
         m[i].display();
      break;

   case 3:
      cout<<"Enter Model: ";
      cin>>search;

      for(int i=0;i<count;i++)
      {
         if(m[i].searchModel(search))
            cout<<"Quantity: "<<m[i].getQuantity()<<endl;
      }
      break;

   case 4:
      cout<<"Enter Model: ";
```

```
            cin>>search;

            for(int i=0;i<count;i++)
            {
               if(m[i].searchModel(search))
                  cout<<"Model Available"<<endl;
            }
            break;
      }

   }while(choice!=5);

   return 0;
}
```

Mobile.h

```
#ifndef MOBILEPHONE_H
#define MOBILEPHONE_H

#include <iostream>
using namespace std;

class MobilePhone
{
private:
   char modelNo[30];
   float price;
   char manufacturer[50];
   int quantity;
```

```
public:
    MobilePhone();
    MobilePhone(char[],float,char[],int);

    void accept();
    void display();

    bool searchModel(char[]);
    int getQuantity();
};

#endif
```

Main.cpp

```
#include "Mobile.h"
#include <cstring>

int main()
{
    MobilePhone m[50];
    int count=0;
    int choice;
    char search[30];

    do
    {
        cout<<"\n1 Add Mobile";
        cout<<"\n2 Display Mobiles";
        cout<<"\n3 Find Quantity of Model";
        cout<<"\n4 Check Availability";
        cout<<"\n5 Exit";
```

```
cout<<"\nEnter Choice: ";
cin>>choice;

switch(choice)
{
   case 1:
      m[count].accept();
      count++;
      break;

   case 2:
      for(int i=0;i<count;i++)
         m[i].display();
      break;

   case 3:
      cout<<"Enter Model: ";
      cin>>search;

      for(int i=0;i<count;i++)
      {
         if(m[i].searchModel(search))
            cout<<"Quantity: "<<m[i].getQuantity()<<endl;
      }
      break;

   case 4:
      cout<<"Enter Model: ";
      cin>>search;
```

```
            for(int i=0;i<count;i++)
            {
                if(m[i].searchModel(search))
                    cout<<"Model Available"<<endl;
            }
            break;
        }

    }while(choice!=5);

    return 0;
}
```

```
PS D:\Sakshi_Kolekar_88\CDAC-PGCP-AC\CPP\SelfLearningAssignment\Mobile> g++ .\main.cpp .\MobilePhone.cpp
PS D:\Sakshi_Kolekar_88\CDAC-PGCP-AC\CPP\SelfLearningAssignment\Mobile> .\a.exe

1 Add Mobile
2 Display Mobiles
3 Find Quantity of Model
4 Check Availability
5 Exit
Enter Choice: 1
Enter Model No: s23
Enter Manufacturer: samsung
Enter Price: 450000
Enter Quantity: 45

1 Add Mobile
2 Display Mobiles
3 Find Quantity of Model
4 Check Availability
5 Exit
Enter Choice:
```

3)paySlips

Main.cpp

```
#include "Payslip.h"

int main()
{
    Payslip p;

    cout<<"Enter Employee Details\n";
    p.accept();
```

```
    cout<<"\nEmployee Payslip\n";
    p.display();

    return 0;
}
```

PaySlips.cpp

```
#include "PaySlip.h"
#include <cstring>

Payslip::Payslip()
{
    empNo = 0;
    DA = 0;
    HRA = 0;
    basicSalary = 0;
    empName[0]='\0';
}

Payslip::Payslip(int e,char n[],float d,float h,float b)
{
    empNo=e;
    DA=d;
    HRA=h;
    basicSalary=b;
    strcpy(empName,n);
}

void Payslip::accept()
{
```

```
    cout<<"Enter Employee No: ";
    cin>>empNo;

    cout<<"Enter Employee Name: ";
    cin>>empName;

    cout<<"Enter Basic Salary: ";
    cin>>basicSalary;

    cout<<"Enter DA: ";
    cin>>DA;

    cout<<"Enter HRA: ";
    cin>>HRA;
}

float Payslip::calculateSalary()
{
    return basicSalary + DA + HRA;
}

void Payslip::display()
{
    cout<<"\nEmployee No: "<<empNo;
    cout<<"\nEmployee Name: "<<empName;
    cout<<"\nBasic Salary: "<<basicSalary;
    cout<<"\nDA: "<<DA;
    cout<<"\nHRA: "<<HRA;
    cout<<"\nTotal Salary: "<<calculateSalary()<<endl;
}
```

PaySlip.h

```
#ifndef PAYSLIP_H
#define PAYSLIP_H

#include <iostream>
using namespace std;

class Payslip
{
private:
    int empNo;
    char empName[50];
    float DA;
    float HRA;
    float basicSalary;

public:
    Payslip();
    Payslip(int,char[],float,float,float);

    void accept();
    void display();
    float calculateSalary();
};
#endif
```

```
PS D:\Sakshi_Kolekar_88\CDAC-PGCP-AC\CPP\SelfLearningAssignment\PaySlips> .\a.exe
Enter Employee Details
Enter Employee No: 23
Enter Employee Name: abc
Enter Basic Salary: 34899
Enter DA: 37.90
Enter HRA: 67994.90

Employee Payslip

Employee No: 23
Employee Name: abc
Basic Salary: 34899
DA: 37.9
HRA: 67994.9
Total Salary: 102932
PS D:\Sakshi_Kolekar_88\CDAC-PGCP-AC\CPP\SelfLearningAssignment\PaySlips>
```